



NARASARAOPETA ENGINEERING COLLEGE (AUTONOMOUS)
DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

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Batch Number	DB-2
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Guide	SK. KHAJA MOHIDDIN BASHA (Asst Prof)
Title	SUMMARIZATION OF TEXT AND IMAGE CAPTIONING IN INFORMATION RETRIEVAL USING DEEP LEARNING.
Domain/Technology	DEEP LEARNING
Base Paper Link	https://ieeexplore.ieee.org/document/9709290
Dataset Link	https://www.kaggle.com/code/dongdori/clip-for-text-to-image-retrieval
Software Requirements	Browser: Any latest browser like Chrome Operating System: Windows 7 Server or later Python (COLAB)
Hardware Requirements	System Type: Intel Core i5 or above RAM: 8 GB Number of cores:5 Number of Threads: 4
Abstract	Automated information retrieval, text summarization, and image captioning present significant challenges in natural language processing (NLP) due to the irregular structure, diversity, and complexity of documents and visual data. Traditional summarization methods heavily depend on handcrafted features, massive labeled datasets, and domain-specific knowledge, often focusing only on narrow sub-domains to enhance efficiency. Addressing these limitations, this paper proposes a novel deep learning (DL)-based framework that unifies information retrieval, text summarization, and image captioning. The proposed model involves three primary processes: information retrieval, template generation, and text summarization. Initially, a Bi-directional Long Short-Term Memory (BiLSTM) network retrieves and encodes semantic information from textual data by processing each

	word into a semantic vector. Subsequently, a deep belief network (DBN) is utilized for text summarization, extracting the essential information while maintaining the original semantic meaning. For image captioning, a combined Convolutional Neural Network (CNN) and Recurrent Neural Network (RNN) approach is adopted, leveraging ResNet50 for feature extraction and BiLSTM for generating natural language descriptions, enhanced further with an attention mechanism to improve contextual relevance. The model's performance was validated using benchmark datasets, including the Gigaword Corpus, DUC Corpus, and the MSCOCO dataset. Experimental results demonstrate that the proposed method significantly outperforms traditional and existing DL-based models in terms of precision, recall, and F-score for text summarization tasks. For image captioning, the system achieved higher BLEU scores, reflecting the generation of accurate and meaningful image descriptions. By integrating semantic feature extraction, template generation, and advanced summarization techniques, the presented model not only improves the accuracy of information retrieval and summarization but also effectively bridges the gap between textual and visual content analysis. Future work aims to further enhance the model's performance through hyperparameter optimization and the integration of more sophisticated attention mechanisms.
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Signature of the student(s)

Signature of the Guide

Signature of the project coordinator